Appendix

Section 3 – Data

TABLE 1 - Data Sources and Summary

Series	*Source & Series ID*	*Frequency*	*Span*	*N*	*Units*	*SA*	*Transformation*	*Accessed*
Average Home Prices	ASPUS - US Census/HUD via FRED	Quarterly	1981Q1 - 2019Q4	156	2017 USD	Yes	Detrended	10/06/2026
Median Home Prices	MSPUS - US Census/HUD via FRED	Quarterly	1981Q1 - 2019Q4	156	2017 USD	Yes	Detrended	10/06/2026
Average - Median Home Prices	Constructed from ASPUS and MSPUS	Quarterly	1981Q1 - 2019Q4	156	Share of Median Price (%)	Yes	Detrended	10/06/2026
Top 10% Income Earners	Realtime Inequality	Quarterly	1981Q1 - 2019Q4	157	Share of Income Earned (%)	No	Diff. (1)	10/07/2026
Housing Inventory	CPS/HVS: Table 8 - US Census Bureau	Quarterly	1981Q1 - 2019Q4	157	Housing Units (millions)	Yes	Diff. (1)	10/21/2026

Section 4 – Stationarity

Average Home Price Figures and Tables

FIGURE 1 - Average Home Prices 1981 - 2019

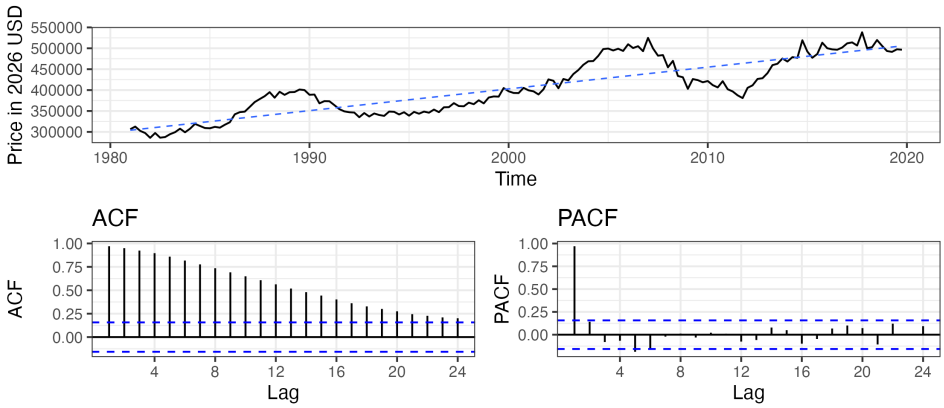


FIGURE 2 - Detreneded Avg. Home Prices

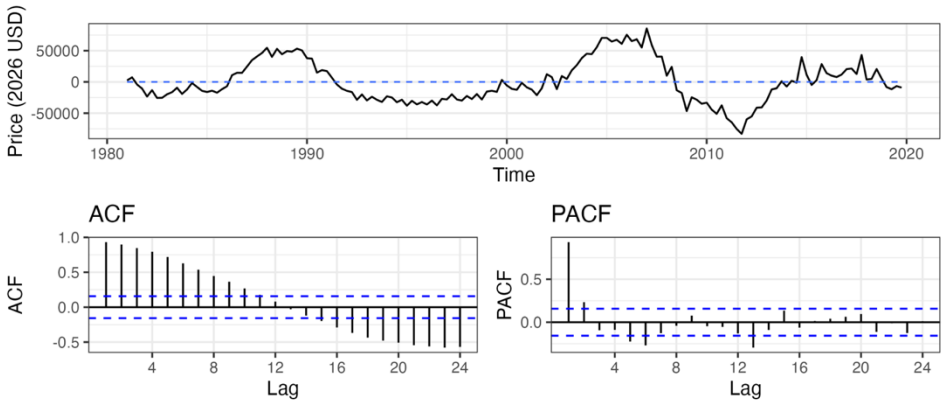


FIGURE 3 - Differenced Avg. Home Prices

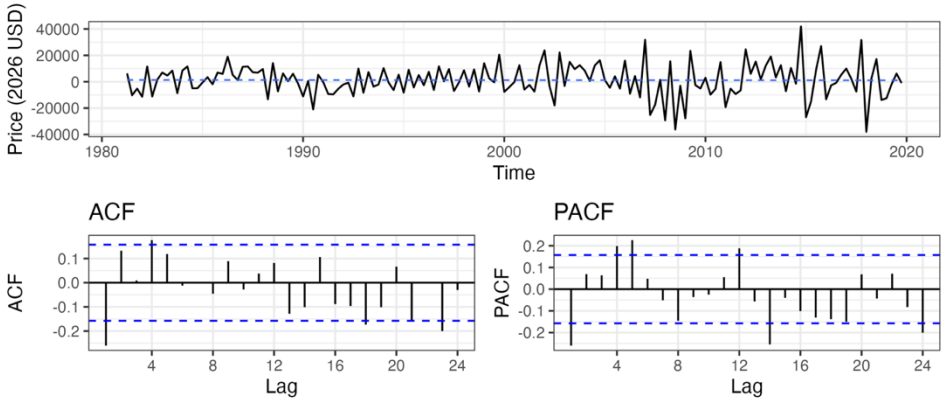


TABLE 2 - Unit-root and stationarity tests

Test	Null	Statistic	Critical values 1%	5%	10%	Decision at 5%
Average Home Prices						
ADF	Unit root	−0.905	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	2.476	0.739	0.463	0.347	Reject null
Detrended Avg. Home Prices						
ADF	Unit root	−1.794	−2.580	−1.950	−1.620	Fail to reject
KPSS	Stationary	0.108	0.739	0.463	0.347	Fail to reject
Differenced Avg. Home Prices						
ADF	Unit root	−9.067	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.059	0.739	0.463	0.347	Fail to reject

Median Home Price Figures and Tables

FIGURE 4 - Median Home Prices 1981 - 2019

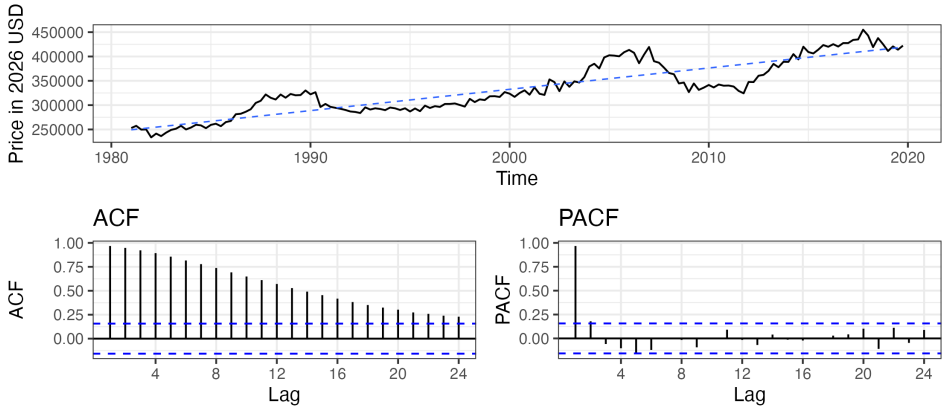


FIGURE 5 - Detrended Med. Home Prices

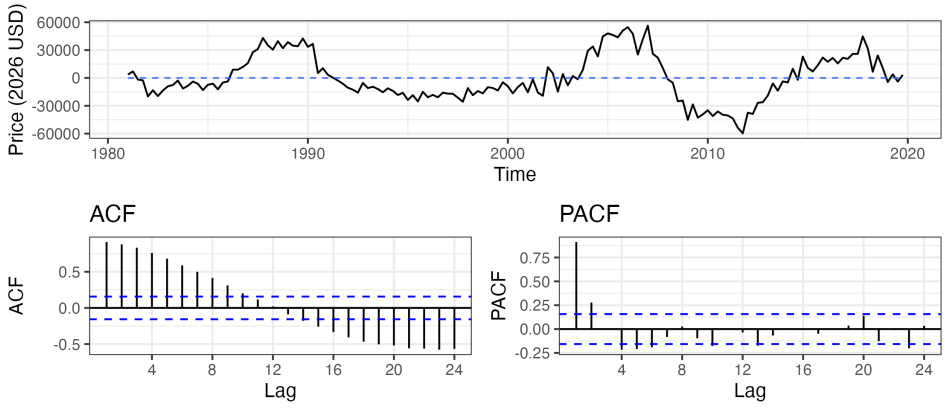


FIGURE 6 - Differenced Median Home Prices 1981 - 2019

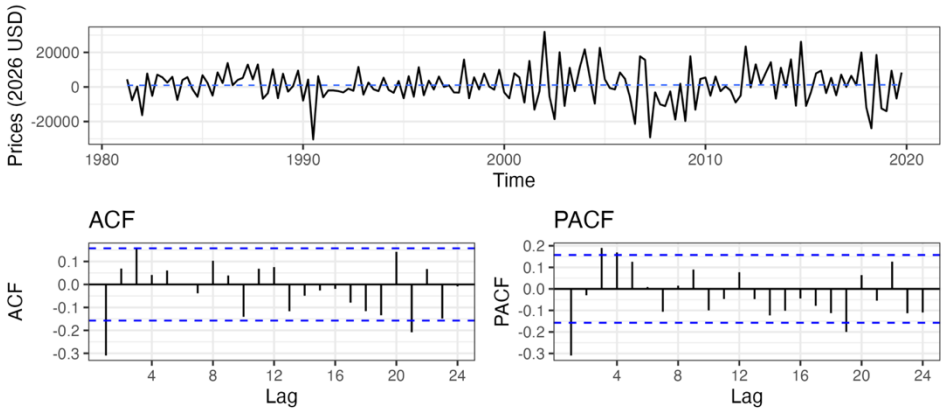


TABLE 3 - ADF and KPSS Test Results of Median Home Prices

			Critical values			
Test	Null	Statistic	1%	5%	10%	Decision at 5%
Median Home Prices						
ADF	Unit root	−0.712	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	2.597	0.739	0.463	0.347	Reject null
Detrended Med. Home Prices						
ADF	Unit root	−1.945	−2.580	−1.950	−1.620	Fail to reject
KPSS	Stationary	0.105	0.739	0.463	0.347	Fail to reject
Differenced Med. Home Prices						
ADF	Unit root	−10.202	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.051	0.739	0.463	0.347	Fail to reject

Home Price Skew Figures and Tables

FIGURE 7 - Skew of Home Price Distribution 1981 - 2019

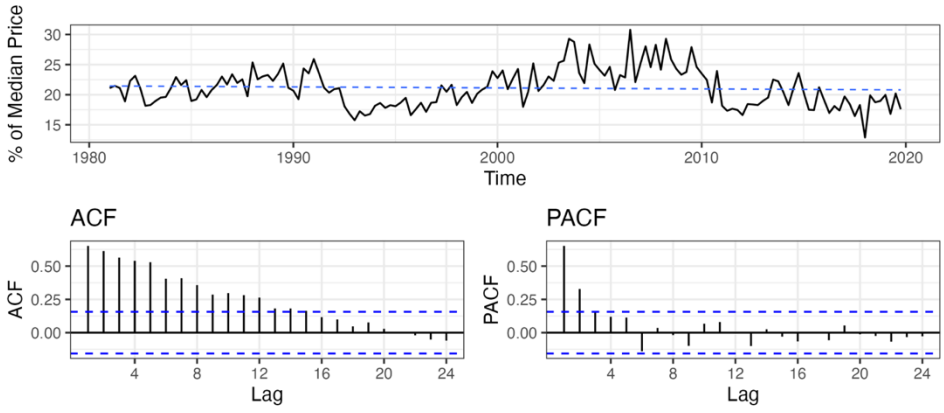


FIGURE 8 - Detrended Skew of Home Prices 1981-2019

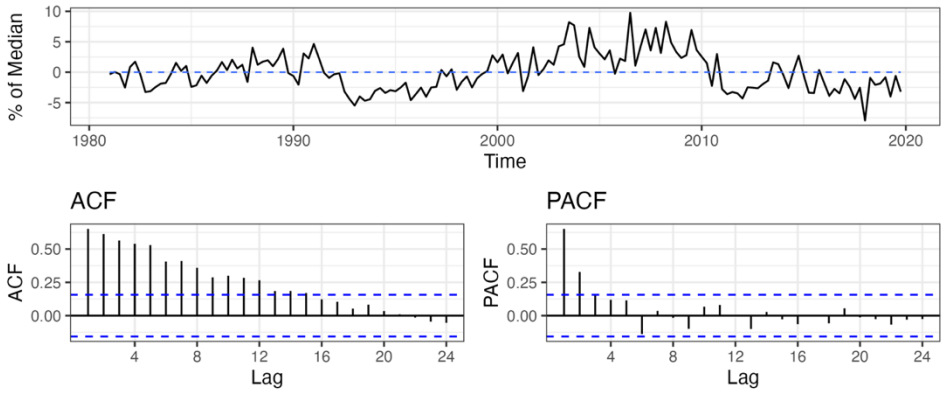


TABLE 4 - ADF and KPSS Test Results of Skew of Home Prices

			Critical values			
Test	Null	Statistic	1%	5%	10%	Decision at 5%
Skew of Home Prices						
ADF	Unit root	−3.569	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.274	0.739	0.463	0.347	Fail to reject
Detrended Skew of Home Prices						
ADF	Unit root	−3.603	−2.580	−1.950	−1.620	Reject null
KPSS	Stationary	0.280	0.739	0.463	0.347	Fail to reject

Income Inequality Figures and Tables

FIGIURE 9 - Income Share of the Top 10% of Income Earners

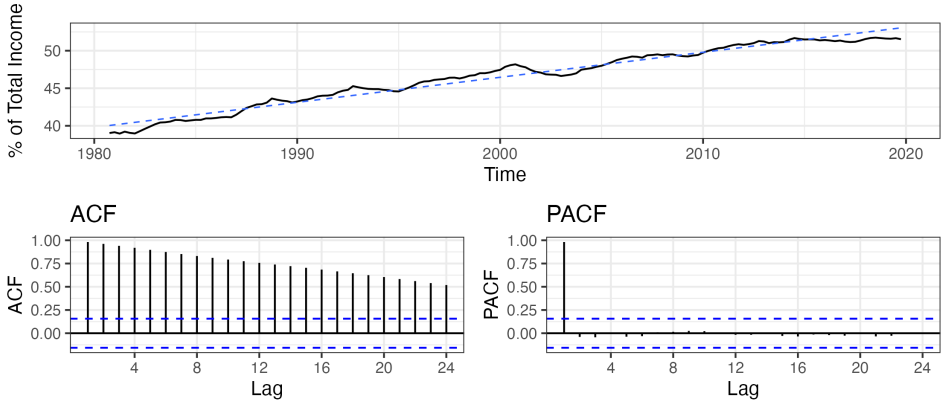


FIGURE 10 - Detrended Income Share of the Top 10% of Income Earners

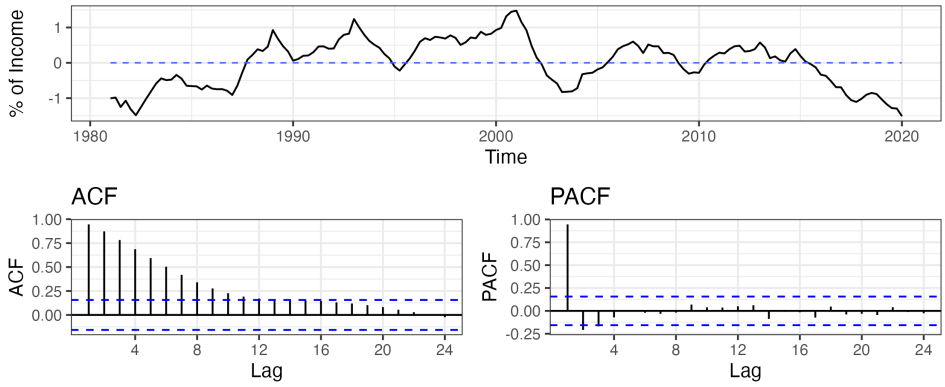


FIGURE 11 - First Differencing of Inequality

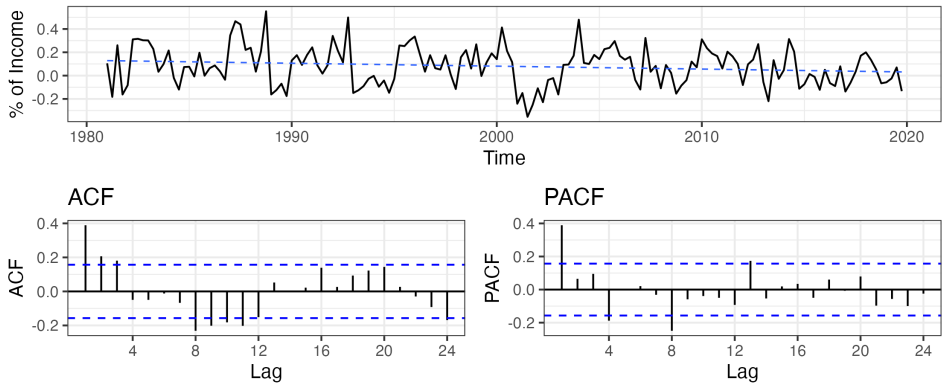


TABLE 5 - ADF and KPSS Test Results of Inequality

			Critical values			
Test	Null	Statistic	1%	5%	10%	Decision at 5%
Inequality						
ADF	Unit root	−1.784	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	3.144	0.739	0.463	0.347	Reject null
Detrended Inequality						
ADF	Unit root	−2.008	−2.580	−1.950	−1.620	Reject null
KPSS	Stationary	0.445	0.739	0.463	0.347	Fail to reject
Differenced Inequality						
ADF	Unit root	−6.358	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.239	0.739	0.463	0.347	Fail to reject

Housing Inventory Figures and Tables

FIGURE 12 - Housing Inventory 1981 - 2019

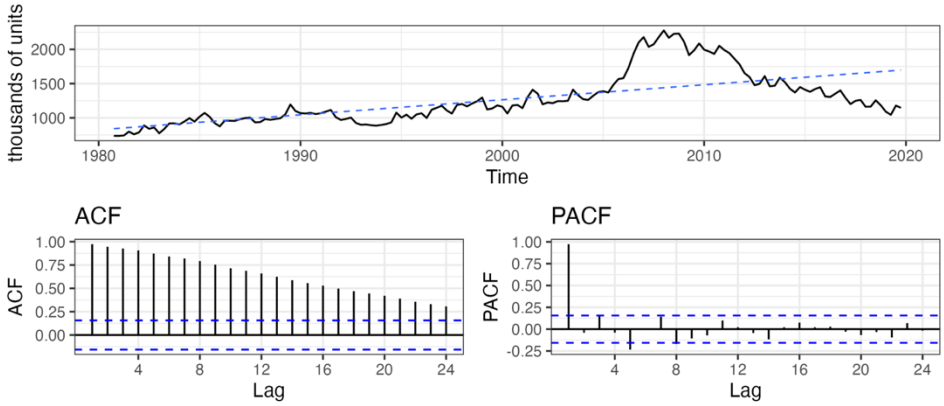


FIGURE 13 - Differenced Housing Inventory

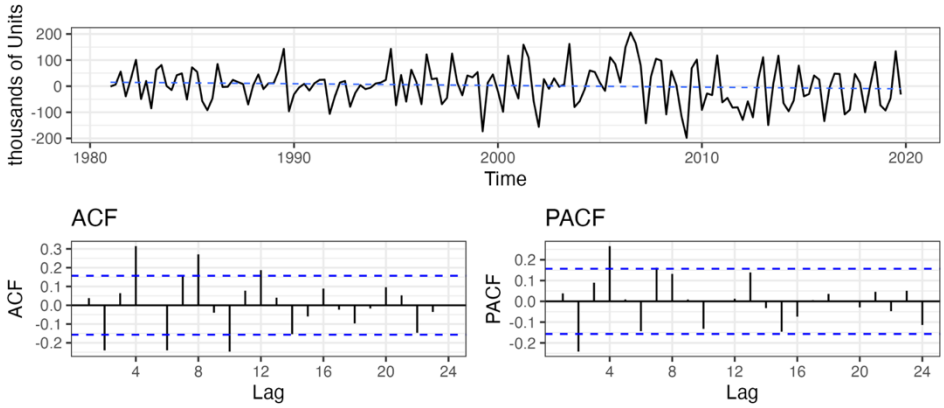


FIGURE 14 - Detrended Housing Inventory

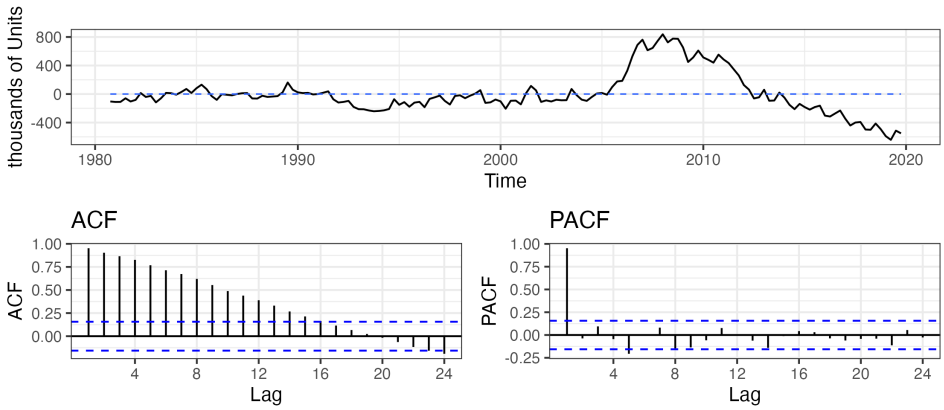


TABLE 6 - ADF and KPSS Test Results of Inventory

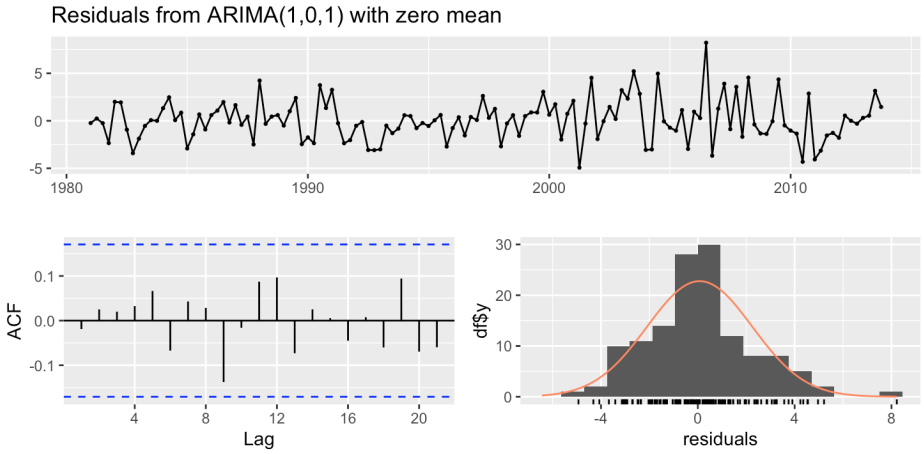
Test	Null	Statistic	Critical values			Decision at 5%
			1%	5%	10%	
Housing Inventory						
ADF	Unit root	−1.671	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	1.800	0.739	0.463	0.347	Reject null
Detrended Inventory						
ADF	Unit root	−1.226	−2.580	−1.950	−1.620	Fail to reject
KPSS	Stationary	0.302	0.739	0.463	0.347	Fail to reject
Differenced Inventory						
ADF	Unit root	−10.920	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.244	0.739	0.463	0.347	Fail to reject

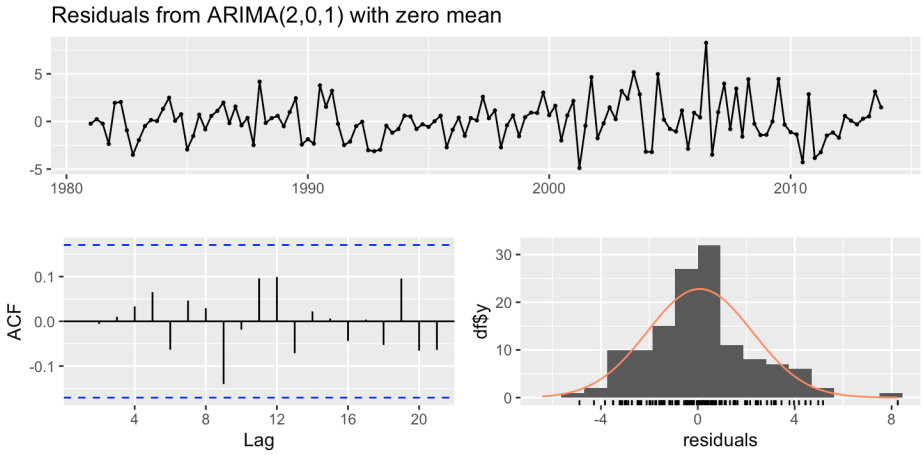
Section 5 – Model Selection

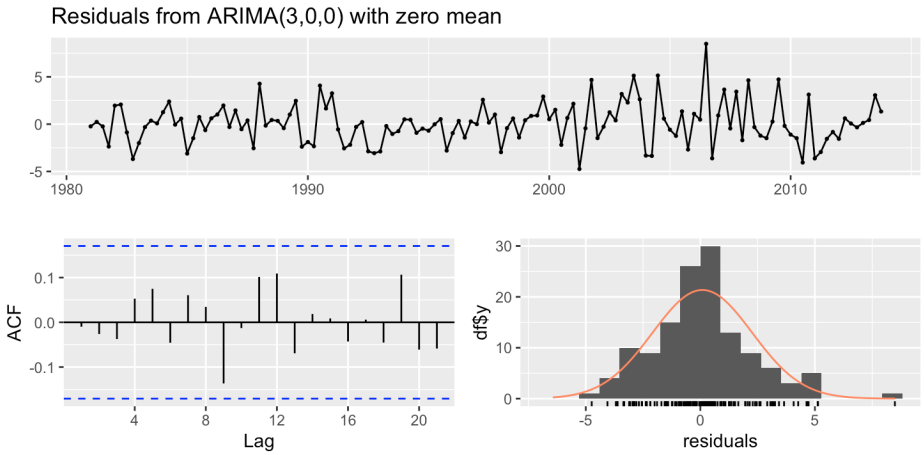
TABLE 7 - Comparison of Home Price Skew Models

Model	k	AIC	AICc	BIC
ARIMA(1,0,0)	2	601.8643	601.9573	607.6299
ARIMA(2,0,0)	3	588.9016	589.0891	597.5500
ARIMA(3,0,0)	4	587.8865	588.2014	599.4177
ARIMA(1,0,1)	3	585.4556	585.6431	594.1040
ARIMA(2,0,1)	4	587.2288	587.5438	598.7600

Section 6 – Serial Correlation

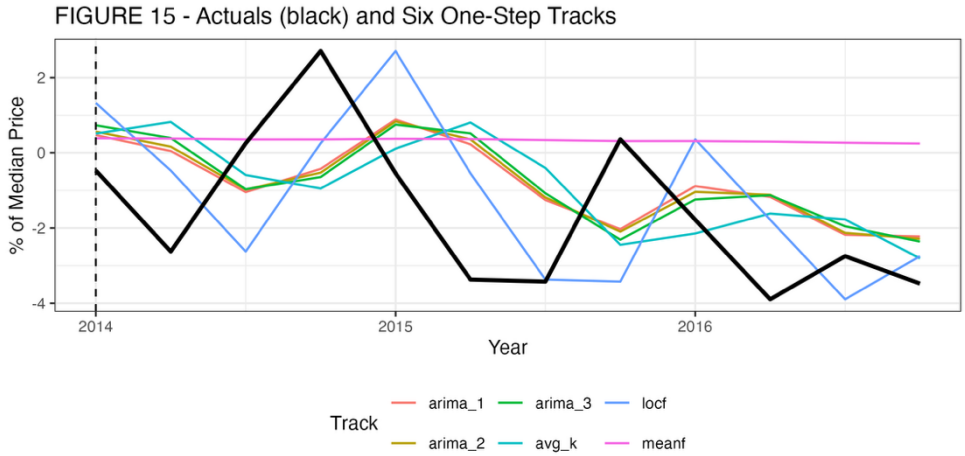






Section 7 - Forecasting

Forecasting Results



Section 8 – Forecast Evaluation

TABLE 8 - Comparison of Models' Loss Functions

Model	MSE	RMSE	MAE
arima_1	4.58521	2.14131	0.78575
arima_2	4.81057	2.19330	0.79825
arima_3	5.20970	2.28248	0.81041
locf	5.49336	2.34379	0.40016
avg_k	5.70716	2.38897	0.71150
meanf	7.43507	2.72673	1.91816